

Graduation Project Progress Report 2

Title	Recommender Systems using generative retrieval based on language models
Summary (in 150 words)	This project investigates how LLM-derived intent, metadata, and recent interaction history can complement collaborative filtering (CF). We implemented a pipeline inspired by IKGR, DynLLM, and CORONA and evaluated it on Goodreads Children, MovieLens 25M, and Amazon Clothing. Temporal evaluations and multi-seed ablations assessed ranking accuracy, long-tail recall, and catalog coverage. Results varied by domain: Goodreads mainly showed long-tail benefits, MovieLens retained LightGCN as the strongest accuracy baseline, and Amazon late fusion modestly improved accuracy and coverage over BPR. Adding intent/metadata or candidate retrieval did not consistently improve performance. We will verify that profiles and intents use only training-period data, revalidate Amazon comparisons, and analyze which users benefit from augmentation. If stable differences emerge, a learned binary gate will select between CF and late fusion using intent, metadata, and recency signals. Controlled intent weighting experiments will test specificity. The goal is reproducible improvement over strong baselines, with domain limitations explicitly reported.
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1. Progress to Date

Since the previous progress report, we have moved from framework design and preliminary IKGR validation to **end-to-end implementation and cross-domain evaluation**. The previous semester concluded with cross-domain exploratory data analysis (EDA), a framework based on IKGR [9], DynLLM [10], and CORONA [6], and a normalized IKGR baseline on Goodreads. An initial heterogeneous knowledge graph (KG) experiment was also completed. DynLLM and CORONA integration, sparse-regime evaluation, and strict cold-start validation remained future work.

The original title reflects our interest in generative retrieval, exemplified by TIGER [3]. The current pipeline evaluates LLM intent extraction, KG/CF fusion, and graph candidate retrieval; autoregressive item-identifier generation is not yet implemented. Related work on semantic late fusion (GRAM [4]), KG retrieval augmentation (K-RagRec [5]), confidence-aware KG augmentation [7], and faithful path explanations [8] provides context for these comparisons. Foundational references include CS224W [1] and Ng's recommendation lectures [2].

During the summer break, we first completed the remaining framework components on Goodreads. We introduced a temporal evaluation protocol and implemented a recency-weighted dynamic user representation inspired by DynLLM. We then implemented and evaluated several CORONA-inspired strategies, including weighted late fusion, graph-based candidate generation, popularity-debiased candidate retrieval, and soft reranking. Rather than assuming that each additional component would improve performance, we conducted systematic ablation studies and reported negative results when a method reduced overall accuracy or long-tail performance.

We also carried out the sparse-regime experiments proposed in the previous report. The Goodreads experiments were extended from the original dense setting ($k=100$) to $k=50$ and $k=30$, and a global temporal evaluation protocol was introduced to examine users with little or no training history. Multi-seed evaluation and deterministic reranking checks were added to improve reproducibility. By mid-July, the Goodreads experiments had therefore expanded from the previously reported IKGR baseline to an evaluation of the full IKGR–DynLLM–CORONA pipeline.

The next stage focused on **cross-domain replication**. Between July and August, the implementation was generalized into a dataset-agnostic pipeline and applied to MovieLens 25M [12], with dataset-specific preprocessing and metadata handling added. The temporal evaluation protocol was also revised to ensure that future interactions did not enter user profiles or knowledge-graph construction. Using a $k=30$ version of MovieLens, we completed both per-user temporal and leakage-safe global temporal experiments, including LLM-based intent extraction, graph construction, collaborative-filtering baselines, three-seed evaluation, and candidate-generation and reranking ablations.

Between August and September, the same framework was evaluated on Amazon Clothing, Shoes and Jewelry [11]. After preparing the dataset, we corrected temporal information leakage by ensuring that **user profiles, intent information, and knowledge-graph components used for training were constructed only from the corresponding training period, without validation- or test-period information**. We then standardized the training configurations and reran the major baseline and ablation experiments. Additional experiments examined the efficiency of LLM-based intent extraction and alternative intent-edge weighting strategies. With this stage completed, the framework had been evaluated across all three domains used in the initial EDA: books, movies, and e-commerce products.

As of October, we have built a **reusable end-to-end experimental pipeline evaluated across three domains**. The evaluation now considers overall recommendation accuracy, long-tail recall, catalog coverage, temporal generalization, and cold-start user subsets, rather than relying on a single aggregate accuracy metric.

The main finding so far is that intent, metadata, and recent interaction history do not produce consistent improvements across all datasets and metrics. Their effects depend strongly on how the signals are integrated: configurations that improve long-tail exposure or catalog coverage may reduce overall ranking accuracy, while fusion strategies that are effective in one domain may not generalize to another. Accordingly, the project objective has evolved from simply demonstrating that LLM-derived intent improves recommendation to identifying **the data conditions and integration strategies under which intent, metadata, and temporal information can effectively complement collaborative filtering**.

2. Preliminary Results

2.1. Cross-domain Evaluation

The completed experiments show that the effects of intent, metadata, recency, and graph-based retrieval vary by dataset and evaluation metric. We therefore assess overall ranking accuracy using NDCG@10, alongside Tail Recall@10 and Coverage@10. Tail Recall measures retrieval of less-popular relevant items under each experiment's popularity definition; Coverage measures the fraction of the catalog appearing in the users' top-10 recommendations. These metrics distinguish ranking gains from broader catalog exposure.

Table 1 summarizes representative configurations and their collaborative-filtering comparisons under per-user temporal evaluation (TO), in which evaluated users have training history. NDCG values are three-seed means with population standard deviations; the remaining metrics are three-seed means. BPR and LightGCN rows use the corresponding recorded TO settings. Dataset scale, filtering, and interaction definitions differ, so comparisons should be made within each dataset rather than between their absolute scores.

Table 1. Representative results under per-user temporal evaluation

Goodreadst[†]: historical profiles; train-only revalidation required.

Dataset	Configuration	NDCG@10 Mean $\pm$ SD	Tail Recall@10	Coverage@10
Goodreads Children [†] (k=30)	BPR	0.0814 $\pm$ 0.0008	0.0110	0.3641
Goodreads Children [†] (k=30)	LightGCN	0.0820 $\pm$ 0.0006	0.0047	0.1757
Goodreads Children [†] (k=30)	KG + recency	0.0822 $\pm$ 0.0005	0.0170	0.5164
MovieLens 25M (k=30)	LightGCN	0.0854 $\pm$ 0.0005	0.0013	0.3366
MovieLens 25M (k=30)	Late fusion	0.0822 $\pm$ 0.0003	0.0033	0.4488
Amazon Clothing	BPR	0.1744 $\pm$ 0.0011	0.0442	0.5722
Amazon Clothing	LightGCN	0.1575 $\pm$ 0.0010	0.0292	0.2159
Amazon Clothing	Late fusion	0.1781 $\pm$ 0.0008	0.0484	0.6383

Notes: Seeds are 2020, 2021, and 2022. SD is population SD (ddof=0), not a confidence interval. Goodreadst[†] results do not establish strict cold-start performance.

2.2. Goodreads: Long-tail Benefits under Sparse Filtering

Goodreads Children [13], [14] was the first test bed for the IKGR-, DynLLM-, and CORONA-inspired components. In the dense $k=100$ setting, collaborative filtering remained strong in overall accuracy, while KG variants mainly improved long-tail exposure and coverage. Experiments at $k=50$ and $k=30$ extended this comparison to broader, less restrictive subsets. Because each filtering threshold changes the user and item population, these runs do not isolate the causal effect of sparsity or establish a monotonic improvement as k decreases.

KG-off is our matrix-factorization ablation trained with the BPR objective; BPR is the separate RecBole baseline used as an implementation check. Under the $k=30$ TO split, KG-off achieved Tail Recall@10 of 0.0105. Intent-plus-metadata KG increased this to 0.0139, and adding recency increased it to 0.0170. The popularity-debiased candidate generator reached 0.0273, but its NDCG@10 fell to 0.0127. The strongest tail result therefore came with a substantial loss in overall ranking accuracy.

KG plus recency achieved NDCG@10 of 0.0822, compared with 0.0814 for BPR and 0.0820 for LightGCN. Its Tail Recall@10 and Coverage@10 were 0.0170 and 0.5164, compared with BPR's 0.0110 and 0.3641. The main observation is comparable overall accuracy with stronger long-tail retrieval and broader exposure, rather than a decisive accuracy advantage over the strongest baseline.

Negative ablation results were also reported. At $k=100$ TO, adding attention to KG plus recency reduced Tail Recall@10 from 0.0110 to 0.0097, and late fusion reduced NDCG@10 from 0.0718 to 0.0678 relative to KG plus recency. The tested soft-reranking settings produced only small changes without establishing a clear overall gain. Together, these results show that additional fusion complexity does not automatically improve recommendation.

2.3. MovieLens: Strong Collaborative Filtering Remains Difficult to Surpass

On the $k=30$ subset of MovieLens 25M under TO, LightGCN achieved the highest overall NDCG@10 of 0.0854. The late-fusion IKGR configuration reached 0.0822, exceeding BPR's 0.0769 but remaining below LightGCN. This differs from Goodreads and Amazon: the additional signals improved over a simpler CF baseline without replacing the strongest accuracy baseline.

The heterogeneous intent-plus-metadata configuration achieved Tail Recall@10 of 0.0041 and Coverage@10 of 0.5142. Soft reranking with $\lambda=0.50$ increased these metrics from 0.0033 to 0.0043 and from 0.4602 to 0.4997 relative to KG plus recency, while NDCG@10 decreased from 0.0796 to 0.0787. This is a measurable accuracy–exposure trade-off.

However, popularity-debiased hard candidate retrieval was a stronger negative result: NDCG@10 fell to 0.0012, and tail recall and coverage also fell below the recency model. Candidate debiasing therefore did not improve long-tail performance in every domain.

An earlier global-temporal implementation constructed user profiles outside the exact global training partition. That evaluation was discarded, and profiles were rebuilt exclusively from the corresponding training interactions. In the final global $k=30$ experiment, LightGCN achieved overall NDCG@10 of 0.0714 and cold0 (users with no training interactions) NDCG@10 of 0.0581, compared with 0.0421 and 0.0244 for KG plus recency. These results provide no evidence of an advantage over the strongest baseline for users reported as having no training history.

The older Goodreads global results used a different profile protocol and are not combined with the corrected MovieLens results as equivalent cold-start evidence. Training-period profile construction also does not by itself establish strict inductive evaluation, since static catalog metadata may remain available.

2.4. Amazon Clothing: Integration Strategy Has a Large Effect

Amazon Clothing showed the clearest case of simultaneous improvement in ranking accuracy and coverage under TO. BPR achieved NDCG@10 of 0.1744 and Coverage@10 of 0.5722, while late fusion reached 0.1781 and 0.6383. Recall@10 also increased from 0.1972 to 0.2029. The NDCG and coverage changes correspond to approximately 2.1% and 11.6% relative increases in the reported means. Late fusion also exceeded LightGCN (NDCG@10 0.1575) in this recorded TO comparison. Section 3 will revalidate the baselines with comparable tuning budgets.

The gain did not extend to every integration strategy. The direct heterogeneous intent-plus-metadata configuration reached only 0.0381 NDCG@10, and KG plus recency reached 0.1287. Late fusion recovered performance to 0.1781. This large variation suggests that the integration method matters as much as the availability of additional information; it does not establish that intent alone caused the improvement. A comparison with intent disabled within the same fusion structure remains necessary to isolate its contribution.

Popularity-debiased candidate retrieval produced the highest Tail Recall@10 among the reported primary TO configurations, at 0.0501. Its NDCG@10 was 0.1530, below both BPR and late fusion. As in Goodreads, this configuration improved long-tail exposure while sacrificing overall accuracy. Soft reranking improved gradually over its recency starting point, but did not match the late-fusion model's accuracy.

Under global temporal evaluation with training-period profiles, the warm-user gain did not transfer to the reported cold0 subset. LightGCN and KG-off achieved cold0 NDCG@10 of 0.0301 and 0.0254, compared with 0.0074 for late fusion. Hard candidate retrieval was especially weak for these users. With candidate pool size $M=500$, candidate Recall@M was about 0.13 under the global split, limiting what the downstream ranker could recover. The current framework therefore cannot be presented as a universal cold-start solution; users without usable history require a separate fallback strategy.

2.5. Overall Interpretation

Across the three datasets, the experiments do not support the simple hypothesis that adding LLM-derived intent, metadata, or recency consistently improves recommendation accuracy. **The highest-accuracy configuration and the configuration with the strongest tail recall or coverage are often different.** Goodreads showed comparable accuracy with stronger long-tail exposure, MovieLens retained LightGCN as the strongest accuracy baseline, and Amazon showed a small mean accuracy gain together with broader coverage under late fusion.

Three conclusions follow within the evaluated settings. KG signals can improve long-tail exposure without improving overall accuracy. Recency and score-level fusion can improve or recover accuracy, but their effects depend on the domain and starting model. Candidate retrieval and reranking can adjust the accuracy–exposure balance, yet hard candidate restriction can also reduce both accuracy and long-tail retrieval. More components or more complex integration do not, by themselves, guarantee better performance.

These findings remain preliminary. Three-seed variation is reported, but statistical significance and the independent contribution of intent have not yet been established. The exact training scope of legacy Goodreads profiles, and the train-only activity and popularity definitions used for cold-user and tail slices, require further audit before final claims are made. The results motivate a more specific research question: **under which data conditions and integration strategies can intent, metadata, and temporal information complement conventional collaborative filtering?**

3. Plan for Final Submission

Based on the current results and our advisor's feedback, the remaining work will investigate **when intent, metadata, and temporal information are beneficial, and whether their selective use can reproducibly improve conventional collaborative filtering**. The cross-domain observations will guide a focused set of analyses and model refinements.

First, we will revalidate the promising **warm-user temporal setting on Amazon Clothing**. NDCG@10 will be the primary criterion, accompanied by Recall@10, Tail Recall@10, and Coverage@10. Late fusion will be compared with BPR, KG-off, and LightGCN using the same temporal split, seeds, and evaluation definitions, with comparable tuning budgets. Results from the other domains will document the limitations and domain dependence of the approach.

Before model selection, we will audit train-only profiles and derived intents, compute user activity and item popularity from the actual training partition, and verify tail-item and cold-user definitions. Model and threshold choices will use validation data; final test results will not be used for further tuning.

Second, we will conduct a **user-level analysis** of the augmented model (late fusion of CF with intent, metadata, and recency signals) and a fixed CF reference. We will report each user's NDCG@10 difference, improvement/tie/degradation rates, and the size of gains and losses. Training-derived explanatory features will include history length, recency, category diversity, and the amount and specificity of intent and metadata. User dependence remains a hypothesis to test; dataset-level averages do not establish it.

On validation data, we will compare the best single model with an oracle that chooses the better model for each user after observing the outcomes. This is a diagnostic upper bound. If the gain is meaningful and user-level patterns are stable and predictable, we will introduce a lightweight **learnable binary gate** to select between the augmented model and CF baseline:

$$\text{Recommendation}(u) = \begin{cases} \text{Augmented Model}(u), & g(z_u) = 1 \\ \text{CF Baseline}(u), & g(z_u) = 0 \end{cases}$$

Here, z_u denotes a representation of user u derived from training information, and g is the gate whose binary output selects which recommender supplies u 's recommendation list.

The underlying recommenders and their embeddings will be fixed while gate labels are obtained from user-level validation performance differences. Gate training and selection will use separate validation user groups or user-level cross-validation, keeping repeated seed observations of the same user together. The gate threshold will be selected using end-to-end NDCG@10 on the gate-selection partition, rather than classifier accuracy alone. Test outcomes will not enter gate features, labels, or threshold selection.

We will compare the gate with **always using CF, always using the augmented model, fixed-weight fusion, and a simple activity-threshold rule**. The gate will be adopted only if it improves on the strongest single-model and simple selection alternatives on validation. Final evaluation will report multi-seed performance, the selection rate, losses from incorrect selection, and a paired confidence interval for the user-level NDCG difference. If the oracle gain or gate generalization is weak, we will prioritize intent weighting and fusion refinement instead of adding classifier complexity.

Third, we will examine **intent specificity**. Frequently occurring intents may provide limited discriminative information, but frequency alone does not establish that they are unhelpful. Within the same fusion structure, we will compare intent-off, uniform intent weights, IDF-based down-weighting, and frequency-based removal. Frequency statistics will use a fixed, explicitly defined information scope, and weighting strengths and removal thresholds will be chosen on validation. Intents whose removal repeatedly benefits the evaluated user groups may be filtered globally; user-dependent effects will instead inform the selection analysis. These controlled ablations will also distinguish the contribution of LLM-derived intent from gains due to model selection or the fusion architecture.

A separate activity-based strategy will remain secondary. History length is already a gate feature and rule baseline. Low-activity users will be distinguished from users with no training history; a separate cold-start model will be considered only if the analysis establishes a clear need.

The final submission will present the validated protocol, user-level analysis, controlled ablations, and baseline comparisons, including negative results. The intended contribution is **to determine whether user-dependent selection of intent, metadata, and temporal information can reproducibly improve the strongest corresponding CF baseline**. This is a testable strategy, not a guaranteed improvement.

4. References

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